Bachmann-Howard from Below

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Mystery of Impredicative Ordinal Analysis

- A lot of ordinal theoretic proof theory ends at Γ_0.
- Good understanding of ordinals up to ϵ_0 (Gentzen) based on Cantor Normal Form.
- Good understanding of adding Veblen function and reaching Γ_0.
- Mystery starts with Bachmann-Howard ordinal and collapsing function, denoting small ordinals by bigger ones, e.g.

$$\psi(\epsilon_{\aleph_1+1})$$

- This talk: Mystery is unnecessary, replace big ordinals by an extension of the Veblen function to nested Klammer symbole:

$$\varphi\begin{pmatrix} \begin{pmatrix} \beta_1^1 \\ \begin{pmatrix} \cdots \\ \cdots \end{pmatrix} \end{pmatrix} & \begin{matrix} \alpha_1 \\ \cdots \\ \cdots \end{matrix} & \begin{pmatrix} \beta_{m_1}^1 \\ \begin{pmatrix} \cdots \\ \cdots \end{pmatrix} \end{pmatrix} & \cdots & \begin{pmatrix} \beta_1^n \\ \begin{pmatrix} \cdots \\ \cdots \end{pmatrix} \end{pmatrix} & \begin{matrix} \alpha_n \\ \cdots \\ \cdots \end{matrix} & \begin{pmatrix} \beta_{m_n}^n \\ \begin{pmatrix} \cdots \\ \cdots \end{pmatrix} \end{pmatrix} \end{pmatrix}$$

Ordinals up to ϵ_0

Beyond ϵ_0

Counting Upwards

0	1	2	3	$\cdots$	
ω	$\omega + 1$	$\omega + 2$	$\omega + 3$	$\cdots$	
$\omega + \omega = \omega \cdot 2$	$\cdots$	$\omega \cdot 3$	$\cdots$		
$\omega \cdot \omega = \omega^2$	$\cdots$	ω^3	$\cdots$	ω^4	$\cdots$
ω^ω	$\cdots$	ω^{ω^ω}	$\cdots$	$\omega^{\omega^{\omega^\omega}}$	$\cdots$

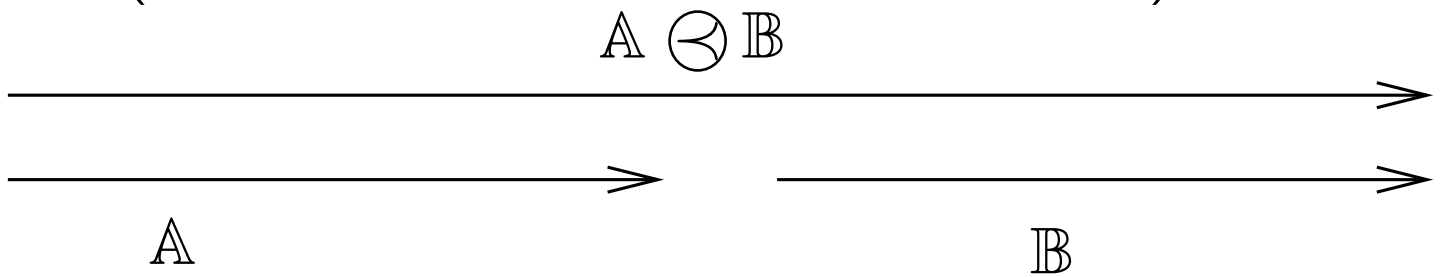
$\epsilon_0 =$ Proof theoretic Strength of PA

An Algebra of Well-Orderings – $\mathbb{N}$, $f^{-1}(\mathbb{A})$, $\mathbb{A} \oplus \mathbb{B}$

- $\mathbb{N} = (\omega, <)$ is a WO.
  - $\mathrm{o}(\mathbb{N}, <) = \omega$.
- $\mathbb{A} = (A, <)$ a well ordering $f : B \to A$ injective, then $f^{-1}(\mathbb{A}) := (B, f^{-1}(<))$ is a well ordering.
- Let for $\mathbb{A}$, $\mathbb{B}$

$$\mathbb{A} \oplus \mathbb{B} := (|\mathbb{A}| + |\mathbb{B}|, \prec)$$

It is a WO (elements of $\mathbb{A}$ ordered below those of $\mathbb{B}$):



We have $\mathrm{o}(\mathbb{A} \oplus \mathbb{B}) = \mathrm{o}(\mathbb{A}) + \mathrm{o}(\mathbb{B})$.

Constructions of Well-Orderings – $\mathbb{A} \cdot \mathbb{B}$, $2^{\mathbb{A}}$

- Let for $\mathbb{A}$, $\mathbb{B}$ WO let

$$\mathbb{A} \cdot \mathbb{B} := (|\mathbb{B}| \times |\mathbb{A}|, \prec_{\text{lex}})$$

Here, $\prec_{\text{lex}}$ = lexicographic ordering on pairs $(b, a) \in |\mathbb{B}| \times |\mathbb{A}|$.
$\mathbb{A} \cdot \mathbb{B}$ is a WO.

$$\mathrm{o}(\mathbb{A} \cdot \mathbb{B}) = \mathrm{o}(\mathbb{A}) \cdot \mathrm{o}(\mathbb{B}) \ .$$

- Let for $\mathbb{A}$ WO

$$|2^{\mathbb{A}}| := \{\langle a_0, \ldots, a_{n-1}\rangle \in |\mathbb{A}|^* \mid a_0 \succ \cdots \succ a_{n-1}\} \ .$$

where $\prec_{2^{\mathbb{A}}}$ lexicographic ordering on descending sequences.
Then $2^{\mathbb{A}}$ is a WO, $\mathrm{o}(2^{\mathbb{A}}) = 2^{\mathrm{o}(\mathbb{A})}$.

Constructions of Well-Orderings – $\mathbb{A}^{\mathbb{B}}$

- We can define $\mathbb{A}^{\mathbb{B}}$

$$\mathrm{o}(\mathbb{A}^{\mathbb{B}}) = \mathrm{o}(\mathbb{A})^{\mathrm{o}(\mathbb{B})} \subseteq 2^{\mathbb{A}\cdot\mathbb{B}}$$

  Essentially forming Cantor Normal Form (CNF) with basis $\gamma := \mathrm{o}(\mathbb{A})$, i.e. forming terms

$$\gamma^{\beta_1} \cdot \alpha_1 + \cdots + \gamma^{\beta_k} \cdot \alpha_k$$

  where β_i are descending elements of $\mathbb{B}$,
  and α_i are non-zero elements of $\mathbb{A}$.

- Define

$$\mathrm{OT}_n := \underbrace{\mathbb{N}^{\mathbb{N}^{\cdots^{\mathbb{O}}}}}_{n \text{ times}} .$$

- Then

$$\mathrm{o}(\mathrm{OT}_n) = \underbrace{\omega^{\omega^{\cdots^{0}}}}_{n \text{ times}} \qquad \epsilon_0 = \sup_{n\in\omega} \mathrm{o}(\mathrm{OT}_n)$$

Ordinals up to ϵ_0

Beyond ϵ_0

Reaching $\varphi_1 0$

- Extend system of CNF, by adding a new ϵ-term whenever we have exhausted CNF

0	$\cdots$	ω^0	$\cdots$	ω^{ω^0}	$\cdots$	$\omega^{\omega^{\omega^0}}$	$\cdots$
ϵ_0	$\epsilon_0 + 1$	$\epsilon_0 + 2$	$\cdots$				
$\epsilon_0 + \omega$	$\cdots$	$\epsilon_0 \cdot 2$	$\cdots$	$\epsilon_0 \cdot 3$	$\cdots$		
$\epsilon_0 \cdot \omega = \omega^{\epsilon_0+1}$	$\cdots$	$\omega^{\omega^{\epsilon_0+1}}$	$\cdots$	$\omega^{\omega^{\omega^{\epsilon_0+1}}}$	$\cdots$		
ϵ_1	$\cdots$	ϵ_2	$\cdots$	ϵ_3	$\cdots$		
ϵ_ω	$\cdots$	ϵ_{ω^ω}	$\cdots$	$\epsilon_{\omega^{\omega^\omega}}$	$\cdots$		
ϵ_{ϵ_0}	$\cdots$	$\epsilon_{\epsilon_{\epsilon_0}}$	$\cdots$	$\epsilon_{\epsilon_{\epsilon_{\epsilon_0}}}$	$\cdots$		
$\varphi_1 0$							

Component Function

- The ordinal notation system is no longer defined in one go (creating finitely nested Cantor Normal Forms).
- Instead
  - We add elements in CNF until we don't get more elements.
  - Then we add ϵ_0, continue with CNF until we don't get more elements etc.
- Every ordinal term is built from other ordinals built before:

$$\mathrm{k} : \mathrm{T} \to \mathcal{P}^{\mathrm{fin}}(\mathrm{T})$$
$$\mathrm{k}(\omega^{\alpha_1} \cdot n_1 + \cdots + \omega^{\alpha_m} \cdot n_m) = \{\alpha_1, \ldots, \alpha_m, n_1 - 1, \ldots, n_m - 1\}$$
$$\mathrm{k}(\epsilon_\alpha) = \{\alpha\}$$

- Every ordinal notation has a finite length

$$\mathrm{length}(\alpha) \in \mathbb{N}$$

- We have a **precedent order** $\prec'$ determining which ordinal to select next.

$\prec'$ for ϵ_α



ϵ_α
ordered by α

$\omega^{\alpha_1} m_1 + \cdots + \omega^{\alpha_k} n_k$
ordered by double lexicographic ordering on descend. sequ. of pairs $\langle(\alpha_1, m_1), \ldots, (\alpha_k, m_k)\rangle$

- **Remark:** $\epsilon_0 + 1 = \omega^{\epsilon_0} + \omega^0 \prec' \epsilon_0$ but $\epsilon_0 \prec \epsilon_0 + 1$.

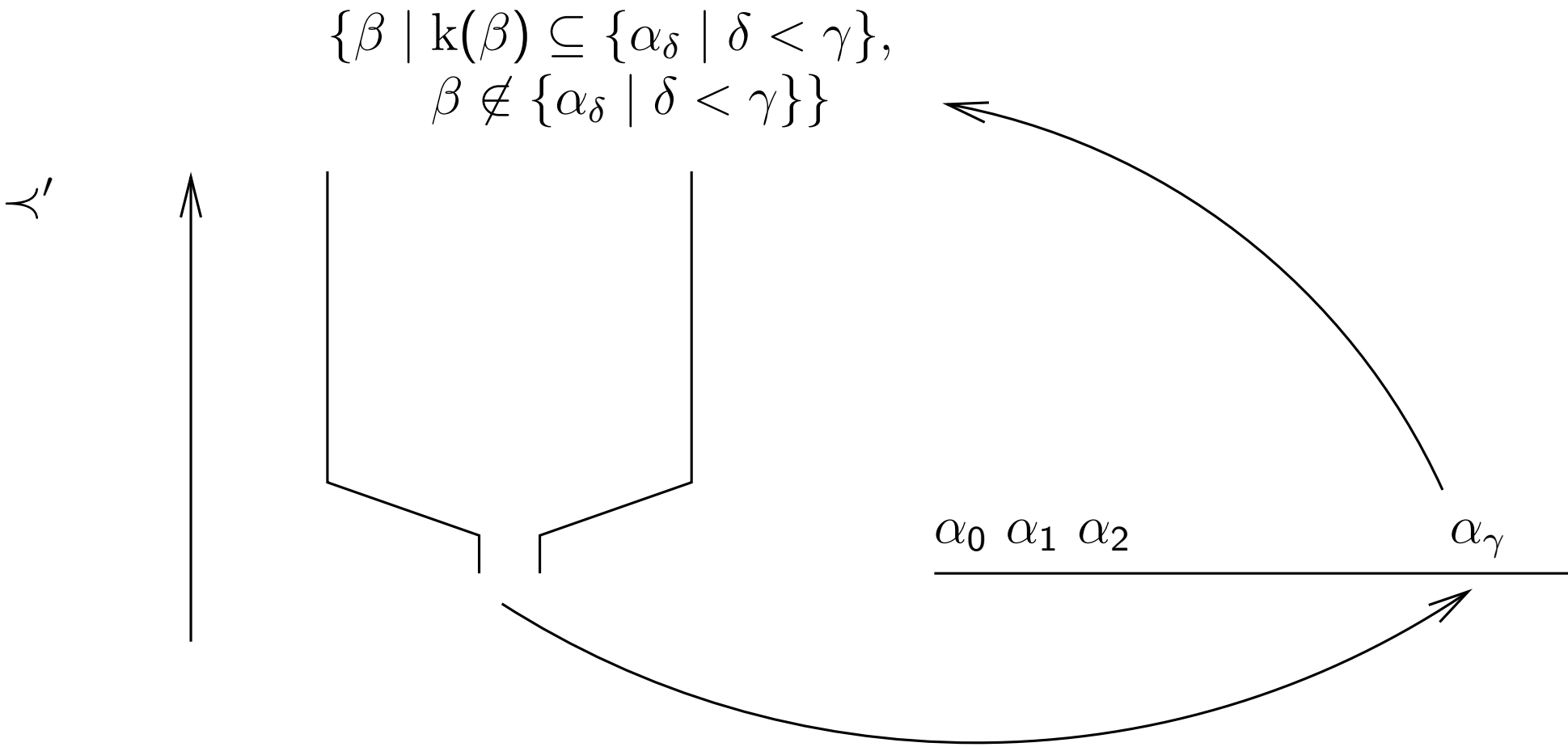
Process of Enumerating Ordinals

- We enumerate the ordinals from below using

$$\begin{aligned} \alpha_\gamma &= \begin{cases} \min{}_{\prec'}\{\delta \in \mathrm{T} \mid \mathrm{k}(\delta) \subseteq \mathrm{A}_\gamma \wedge \delta \notin \mathrm{A}_\gamma\}, & \text{if min exists,} \\ \bot, & \text{otherwise.} \end{cases} \\ \mathrm{A}_\gamma &= \{\alpha_\rho \mid \rho < \gamma\} \end{aligned}$$

- We want that
  - $\alpha_\gamma \downarrow$ unless $\mathrm{A}_\gamma = \mathrm{T}$ ($\mathrm{T} =$ the set of ordinal notations).
  - (α_γ) is an increasing sequence of ordinal notations with no gaps.

Process of Introducing Ordinals



Achieving (α_γ) Increasing

- In order to achieve that this process is increasing we need

$$\text{(OS 3)} \qquad \alpha \prec \beta \to (\alpha \preceq \mathrm{k}(\beta) \vee \alpha \prec' \beta)$$

  - The reason for selecting α before β when enumerating them in increasing order is that
    either β has components which were not available when selecting α,
    or $\alpha \prec' \beta$.

- Furthermore, before we can introduce an ordinal, we must introduce its components

$$\text{(OS 1)} \qquad \mathrm{k}(\alpha) \prec \alpha$$

- Components of an ordinal notation form part of the ordinal notation, therefore

$$\text{(OS 2)}\ \mathrm{length}[\mathrm{k}(\alpha)] < \mathrm{length}(\alpha)$$

Achieving Existence of min

- In order to achieve that we can define

$$\min{}_{\prec'}\{\beta \mid \mathrm{k}(\beta) \subseteq \mathrm{A}_\gamma \wedge \beta \notin \mathrm{A}_\gamma\}$$

(unless $\mathrm{A}_\gamma = \mathrm{T}$) we require

(OS 4) If $A \subseteq \mathrm{T}$ is $\prec$-WO, then $\mathrm{T} \restriction A := \{\alpha \in \mathrm{T} \mid \mathrm{k}(\alpha) \subseteq A\}$ is $\prec'$-WO.

- Then the min can be found:
  - By IH A_γ is WO (since α_γ chosen in increasing order).
  - Therefore, the set, $\min_{\prec'}$ is chosen from, is $\prec'$-WO

Definition Ordinal System

- An ordinal system (OS) is a tuple

$$(\mathrm{T}, \prec, \prec', \mathrm{k}, \mathrm{length})$$

where

- T is a set
- $\prec'$, $\prec$ are linear orderings on T,
- $\mathrm{k} : \mathrm{T} \to_\omega \mathrm{T}$,
- $\mathrm{length} : \mathrm{T} \to \mathbb{N}$,

s.t. for all $\alpha, \beta \in \mathrm{T}$ the following holds:

(OS 1) $\mathrm{k}(\alpha) \prec \alpha$,

(OS 2) $\mathrm{length}(\mathrm{k}(\alpha)) < \mathrm{length}(\alpha)$

(OS 3) $\alpha \prec \beta \to (\alpha \preceq \mathrm{k}(\beta) \vee \alpha \prec' \beta)$

(OS 4) If $A \subseteq \mathrm{T}$ is $\prec$-WO, then $\mathrm{T} \restriction A$ is $\prec'$-WO.

- Ordinal system is elementary if (OS 4) provable in PRA.
- If T is an elementary ordinal systems, $\mathrm{ID}_1 \vdash \mathrm{WO}(T)$.

$\prec'$ for Veblen Function



$\varphi_\alpha\beta$
ordered lexicographically by (α, β)

$\omega^{\alpha_1} m_1 + \cdots + \omega^{\alpha_k} n_k$

$\prec'$ for n-ary Veblen Function



$\varphi_n(\alpha_1, \ldots, \alpha_n)$
ordered lexicographically by
$(n, \alpha_1, \ldots, \alpha_n)$

$\omega^{\alpha_1} m_1 + \cdots + \omega^{\alpha_k} n_k$

Schütte Klammer Symbole

- Once φ-function is exhausted use a φ-function with infinitely many arguments.
  - arguments indexed by ordinals,
  - but only finitely many $\neq 0$.
- So we have

$$\begin{array}{lccccccccccccccc} \varphi(& 0 & \cdots & 0 & \alpha_0 & 0 & \cdots & 0 & \alpha_1 & 0 & \cdots & 0 & \alpha_n & 0 & \cdots & 0) \\ & & & & \uparrow & & & & \uparrow & & & & \uparrow & & & \\ & & & & \beta_0 & & & & \beta_1 & & & & \beta_n & & & \end{array}$$

- Written as

$$\varphi \begin{pmatrix} \alpha_0 & \alpha_1 & \cdots & \alpha_n \\ \beta_0 & \beta_1 & \cdots & \beta_n \end{pmatrix}$$

(where $\alpha_i \neq 0$).

$\prec'$ for Schütte Klammer Symbole



$$\varphi \begin{pmatrix} \alpha_1 & \cdots & \alpha_n \\ \beta_1 & \dots & \beta_n \end{pmatrix}$$

ordered double lexicographically by

$((\beta_1, \alpha_1), \dots, (\beta_n, \alpha_n))$

$$\omega^{\alpha_1} m_1 + \cdots + \omega^{\alpha_k} n_k$$

$\prec'$ for Nested Klammer Symbole (AS)



$$\varphi\left(\begin{array}{ccc} & \alpha_1 & \\ \left(\begin{array}{ccc} \beta_1^1 & \cdots & \beta_{m_1}^1 \\ \gamma_1^1 & \cdots & \gamma_{m_1}^1 \end{array}\right) & \cdots & \left(\begin{array}{ccc} \beta_1^n & \cdots & \beta_{m_n}^n \\ \gamma_1^n & \cdots & \gamma_{m_n}^n \end{array}\right) \end{array}\right)$$

ordered double lexicographically by

$$\begin{array}{ll} \langle & (\langle(\gamma_1^1,\beta_1^1),\ldots,(\gamma_{m_1}^1,\beta_{m_1}^1)\rangle,\alpha_1) \\ & ,\ldots, \\ & (\langle(\gamma_1^n,\beta_1^n),\ldots,(\gamma_{m_n}^n,\beta_{m_n}^n),\alpha_n\rangle) \quad \rangle \end{array}$$

$$\omega^{\alpha_1} m_1 + \cdots + \omega^{\alpha_k} n_k$$

Reaching Bachmann-Howard

- We can nest this further:

$$\varphi\left(\begin{array}{ccccc} & & \alpha_1 & & \\ \left(\begin{array}{c}\beta_1^1\\ \left(\begin{array}{c}\cdots\\ \cdots\end{array}\right)\end{array}\right) & \cdots & \left(\begin{array}{c}\beta_{m_1}^1\\ \left(\begin{array}{c}\cdots\\ \cdots\end{array}\right)\end{array}\right) & \cdots & \left(\begin{array}{c}\beta_1^n\\ \left(\begin{array}{c}\cdots\\ \cdots\end{array}\right)\end{array}\right) \end{array}\right)$$

- Using nestings of arbitrary depth one reaches the Bachmann-Howard ordinal $= |\mathrm{ID}_1|$.

Conclusion

- Problem of understanding ordinal notation systems based on collapsing functions.
- Can be overcome by extending the principle of the Veblen function first to n-ary, the Schütte Klammersymbole and then extended Klammer symbole.
- Algebraic structure.
- Relationship to Girard's dilators to be explored.
- Approach has been extended up to the proof theoretic strength of Mahlo.

Optional: Well-ordering prooof for Ordinal Systems

- Let Acc be the accessible part, largest initial segment which is wellfounded.
  - Inductively defined by $\forall x.(\forall y \prec x.y \in \mathrm{Acc}) \to x \in \mathrm{Acc}$.
- Let $\mathrm{Acc}' := \{x \mid \mathrm{k}(x) \subseteq \mathrm{Acc}\}$.
  Since Acc is well founded, so is Acc'.
- We show $\forall x \in \mathrm{Acc}'.x \in \mathrm{Acc}$ by induction on Acc'.
  Then we get by induction on $\mathrm{length}(x)$ $\forall x.x \in \mathrm{Acc}$.
  Assume $x \in \mathrm{Acc}'$.
  - We show $\forall y \prec x.y \in \mathrm{Acc}$ by side inducton on $\mathrm{length}(y)$.
    Then we get $x \in \mathrm{Acc}$.
    So assume $y \prec x$.
  - If $y \preceq \mathrm{k}(x) \subseteq \mathrm{Acc}$ then $y \in \mathrm{Acc}$.
  - Otherwise $y \prec' x$
    and we have $\mathrm{k}(y) \prec y \prec x$,
    by side-IH $\mathrm{k}(y) \subseteq \mathrm{Acc}$,
    $y \in \mathrm{Acc}'$,
    by main-IH $y \in \mathrm{Acc}$.